

RNA junction and register choices in a computational resource for NR4A3 fusion gapmer design

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Abstract

The sequence of a fusion-junction oligonucleotide depends on both the exact RNA join and its position across that join. A primary catalogue of 190 gapmer designs across 38 modelled NR4A3 exon-3 fusion junctions records these choices and their sequence matches. A one-base shift at a TAF15 exon-6/NR4A3 exon-3 junction changes the longest mature-parent duplex spanning the DNA gap from nine base pairs with TFG to eleven with NR4A3. Changing the NR4A3 acceptor from exon 3 to exon 2 produces a different design despite the same fusion-gene names. Under an adopted ten-base criterion, 87 designs (45.8%) meet the parent-pairing threshold; artificial exon joins give a similar 40.6%, so no EMC-specific excess is established. Published patient-derived model descriptions do not resolve their nucleotide-level correspondence to the catalogue. The resource documents conditional design choices and match records. No reagent was synthesised or tested, and sequence criteria do not establish cleavage or biological selectivity.

Keywords: Extraskeletal myxoid chondrosarcoma; NR4A3; fusion junction; antisense oligonucleotide; gapmer; sequence register

1. Introduction

A fusion-junction antisense oligonucleotide must match the exact RNA join. Its position across that join determines how much sequence remains shared with the normal parent transcripts. Extraskelatal myxoid chondrosarcoma (EMC), which commonly carries EWSR1::NR4A3 or TAF15::NR4A3, provides a setting in which these choices can be recorded explicitly.[1,2] Junction-directed antisense oligonucleotides have been studied in Ewing sarcoma.[3] This resource applies established sequence-screening concepts to NR4A3 fusions, asking how junction identity and oligonucleotide register change the matches encountered during design.

2. Methods

The primary catalogue contains 190 designs: five registers across each of 38 modelled in-frame junctions joining five fusion partners to NR4A3 exon 3. Canonical transcripts were obtained from Ensembl.[4] Each design is a 16-nucleotide gapmer with five locked-nucleic-acid residues at either end and a central six-nucleotide DNA gap, with a phosphorothioate backbone throughout. The breakpoint remains inside the DNA gap as the design moves across the junction. Six nucleotides is the cited minimum DNA-gap length, rather than the preferred working length.[5] The machine-readable catalogue preserves sequence, chemistry, junction and screen results.

Five complementary screens describe matches in RefSeq RNA, an exhaustive transcript substitution search, unspliced parent RNA, six mature normal parent transcripts and unambiguous GRCh38 sequence. A near-match pairs at least 14 of the 16 positions. A separate parent-pairing criterion requires at least ten consecutive base pairs spanning the entire DNA gap. This describes total contiguous pairing, not the length of the DNA

segment. Ten base pairs was an adopted criterion, not an experimentally measured cleavage threshold. Per-design match lengths retain the information needed to examine alternative cutoffs.

3. Results

Table 1 illustrates the effects of register and acceptor choice. The centred EWSR1 exon-12/NR4A3 exon-3 and TAF15 exon-6/NR4A3 exon-3 designs have longest mature-parent duplexes spanning the whole gap of eight and nine base pairs, respectively, both with TFG. Neither reaches the ten-base criterion, but both reach an eight-base criterion. Moving the TAF15 design by one base permits eleven consecutive base pairs with normal NR4A3 through the whole gap. This changes the sequence classification; whether it affects cleavage or selectivity remains an experimental question.

Acceptor identity matters just as directly. A separately modelled TAF15 exon-6/NR4A3 exon-2 join produces a different centred sequence and longest matching parent despite identical fusion-gene names and donor-exon labels. Exons are numbered from the transcript's 5' end, including non-coding exons. The two published patient-derived EMC cell models report NR4A3 exon-2 acceptors without a nucleotide boundary or transcript accession sufficient to resolve correspondence to this catalogue.[6] A fusion name or exon label alone therefore does not identify the appropriate sequence. Published engineered constructs provide separately defined exon-3 junctions.[7]

Across the primary catalogue, 87 of 190 designs (45.8%) meet the ten-base mature-parent criterion. Selecting within each junction identifies at least one design below that criterion for 35 of 38 junctions, including all five with published exon-resolved breakpoints. The counts fall to 31 and 23 junctions at nine- and eight-base cutoffs; only

three and two published junctions, respectively, retain such a design. These summarize modelled sequences, not the proportion of patients who could receive a selective reagent.

Artificial exon joins provide a counterpoint: 40.6% meet the same criterion, compared with 45.8% in the EMC panel. Most joins in both sets have not been reported in patients, and the comparison does not establish an EMC-specific excess of parent pairing. The useful output is the conditional record for each junction and register. Complete cutoff comparisons, transcript and genome results, screened scramble controls and model outputs accompany the catalogue and extended evidence.

4. Discussion

The catalogue documents design choices but does not select validated leads. Partial complementarity and gapmer off-target activity are established subjects of study.[8,9] Sequence complementarity and modelled duplex energy do not establish cleavage; the proposed modified chemistry is not captured by the unmodified DNA:RNA temperature calculations.[10] The screens have different search spaces and thresholds, and transcript-alignment outputs were incomplete. Most junctions remain modelled, and no reagent was synthesised or tested. Potency, delivery, safety and a therapeutic window remain unmeasured. Within these limits, the resource connects a confirmed RNA junction to a specified design and its sequence-match record.

Data availability

The accompanying machine-readable sequence catalogue supplies the reagent specifications and match records. Extended evidence retains detailed methods, model outputs, controls and alternative cutoffs; the correction note explains the temperature-

model and cell-model limitations. Code and per-design outputs from the preceding analysis are archived at <https://doi.org/10.5281/zenodo.22229096>. That deposit is not claimed to contain the subsequent interpretive corrections; the accompanying files supply those qualifications.

Declarations

No external funding supported this work. The author has no competing financial interests and declares a nonfinancial interest as an EMC survivor. The study used public data and reports no new recruitment, specimen collection or experiments.

Tristan D. McRae is the sole author and directed the work. Claude (Anthropic) assisted with the original analysis code, sequence screens, literature retrieval and checking, and manuscript drafting. OpenAI models assisted with revisions and checks against repository artifacts; OpenAI Codex assisted with this shorter presentation. The author is responsible for the content; AI assistance does not constitute independent scientific validation.

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Table 1

Junction and register effects on mature-parent pairing. Sequences are written 5' to 3'. Each design has 5–6–5 LNA–DNA–LNA geometry and a phosphorothioate backbone. Exon numbering includes non-coding exons. The duplex length is consecutive pairing with a mature normal parent through all six DNA-gap positions. A value below ten is below the adopted sequence cutoff; it is not a biological selectivity verdict. The exon-2 example lies outside the 190-design primary panel.

Junction and register	Sequence	Longest full-gap parent duplex	Meets ≥10-bp criterion
EWSR1 e12::NR4A3 e3, centred	GGGCATATCATCAAAC	8 bp, TFG	No
TAF15 e6::NR4A3 e3, centred	GGGCATATCTTGTGTG	9 bp, TFG	No
TAF15 e6::NR4A3 e3, adjacent	AGGGCATATCTTGTGT	11 bp, NR4A3	Yes
TAF15 e6::NR4A3 e2, centred	AGTGGGCTCTTGTGTG	9 bp, NR4A3	No